

Hojas de datos

74LS04

Recommended Operating Conditions

Symbol	Parameter	Min	Nom	Max	Units
V_{CC}	Supply Voltage	4.75	5	5.25	V
V_{IH}	HIGH Level Input Voltage	2			V
V_{IL}	LOW Level Input Voltage			0.8	V
I_{OH}	HIGH Level Output Current			-0.4	mA
I_{OL}	LOW Level Output Current			8	mA
T_A	Free Air Operating Temperature	0		70	°C

Electrical Characteristics

over recommended operating free air temperature range (unless otherwise noted)

Symbol	Parameter	Conditions	Min	Typ (Note 2)	Max	Units
V_I	Input Clamp Voltage	$V_{CC} = \text{Min}$, $I_I = -18 \text{ mA}$			-1.5	V
V_{OH}	HIGH Level Output Voltage	$V_{CC} = \text{Min}$, $I_{OH} = \text{Max}$, $V_{IL} = \text{Max}$	2.7	3.4		V
V_{OL}	LOW Level Output Voltage	$V_{CC} = \text{Min}$, $I_{OL} = \text{Max}$, $V_{IH} = \text{Min}$		0.35	0.5	V
		$I_{OL} = 4 \text{ mA}$, $V_{CC} = \text{Min}$		0.25	0.4	
I_I	Input Current @ Max Input Voltage	$V_{CC} = \text{Max}$, $V_I = 7 \text{ V}$			0.1	mA
I_{IH}	HIGH Level Input Current	$V_{CC} = \text{Max}$, $V_I = 2.7 \text{ V}$			20	μA
I_{IL}	LOW Level Input Current	$V_{CC} = \text{Max}$, $V_I = 0.4 \text{ V}$			-0.36	mA
I_{OS}	Short Circuit Output Current	$V_{CC} = \text{Max}$ (Note 3)	-20		-100	mA
I_{CCH}	Supply Current with Outputs HIGH	$V_{CC} = \text{Max}$		1.2	2.4	mA
I_{CCL}	Supply Current with Outputs LOW	$V_{CC} = \text{Max}$		3.6	6.6	mA

Note 2: All typicals are at $V_{CC} = 5 \text{ V}$, $T_A = 25^\circ \text{C}$.

Note 3: Not more than one output should be shorted at a time, and the duration should not exceed one second.

Switching Characteristics

at $V_{CC} = 5 \text{ V}$ and $T_A = 25^\circ \text{C}$

		$R_L = 2 \text{ k}\Omega$				Units
Symbol	Parameter	$C_L = 15 \text{ pF}$		$C_L = 50 \text{ pF}$		
		Min	Max	Min	Max	
t_{PLH}	Propagation Delay Time LOW-to-HIGH Level Output	3	10	4	15	ns
t_{PHL}	Propagation Delay Time HIGH-to-LOW Level Output	3	10	4	15	ns

datasheet: <https://www.alldatasheet.es/datasheet-pdf/view/51022/FAIRCHILD/74LS04.html>

74LS04

■ RECOMMENDED OPERATING CONDITIONS ($V_{SS}=0\text{V}$)

PARAMETER	SYMBOL	RATINGS	UNIT
DC Supply Voltage	V_{DD}	3 ~ 15	V
Input Voltage	V_{IN}	0 ~ V_{DD}	V
Operating Temperature Range	T_a	-40 ~ +85	°C

■ **DC ELECTRICAL CHARACTERISTICS** ($V_{SS}=0V$, $T_a=25^\circ C$, unless otherwise specified.)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT
Quiescent Device Current	I_{DD}	$V_{DD}=5V$, $V_{IN}=V_{DD}$ or V_{SS} $V_{DD}=10V$, $V_{IN}=V_{DD}$ or V_{SS} $V_{DD}=15V$, $V_{IN}=V_{DD}$ or V_{SS}			1.0 2.0 4.0	μA
Low Level Output Voltage	V_{OL}	$ I_O < 1\mu A$ $V_{DD}=5V$ $V_{DD}=10V$ $V_{DD}=15V$		0 0 0	0.05 0.05 0.05	V
High Level Output Voltage	V_{OH}	$ I_O < 1\mu A$ $V_{DD}=5V$ $V_{DD}=10V$ $V_{DD}=15V$	4.95 9.95 14.95			V
Low Level Input Voltage	V_{IL}	$ I_O < 1\mu A$ $V_{DD}=5V$, $V_{OUT}=4.5V$ $V_{DD}=10V$, $V_{OUT}=9V$ $V_{DD}=15V$, $V_{OUT}=13.5V$			1.0 2.0 3.0	V
High Level Input Voltage	V_{IH}	$ I_O < 1\mu A$ $V_{DD}=5V$, $V_{OUT}=0.5V$ $V_{DD}=10V$, $V_{OUT}=1V$ $V_{DD}=15V$, $V_{OUT}=1.5V$	4.0 8.0 12.0			V
Low Level Output Current (Note 3)	I_{OL}	$V_{DD}=5V$, $V_{OUT}=0.4V$ $V_{DD}=10V$, $V_{OUT}=0.5V$ $V_{DD}=15V$, $V_{OUT}=1.5V$	0.44 1.1 3.0	0.88 2.25 8.8		mA
High Level Output Current (Note 3)	I_{OH}	$V_{DD}=5V$, $V_{OUT}=4.6V$ $V_{DD}=10V$, $V_{OUT}=9.5V$ $V_{DD}=15V$, $V_{OUT}=13.5V$	-0.44 -1.1 -3.0	-0.88 -2.25 -8.8		mA
Input Current	I_{IN}	$V_{DD}=15V$, $V_{IN}=0V$ $V_{DD}=15V$, $V_{IN}=15V$		-10^{-5} 10^{-5}	-0.30 0.30	μA

datasheet: <https://www.alldatasheet.es/datasheet-pdf/view/172773/UTC/CD4069.html>

4. Etapa de Simulación

- Función de transferencia V_{out} vs V_{in} (TTL y CMOS superpuestas) →

TTL

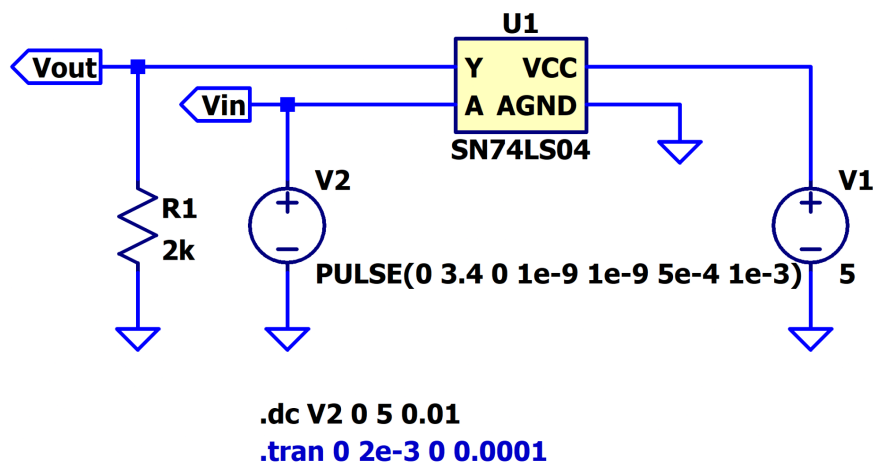


imagen 1. Simulación del circuito compuerta logica SN74LS04

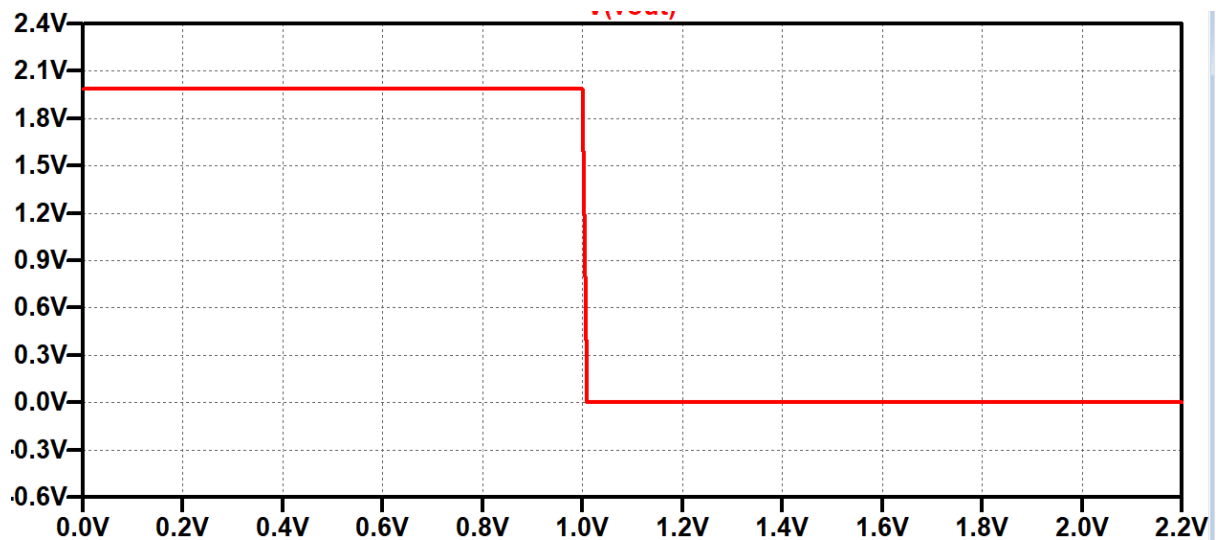


imagen 2. Función de transferencia V_{out} vs V_{in}

- Respuesta en frecuencia a 1kHz, 10kHz, 100kHz, 500kHz y 1MHz → **"Ver Figura 2 en anexo"**
- Consumo de potencia simulado (comparar con los cálculos de la sección 3.2/3.3) → **"Ver Figura 3 en anexo"**

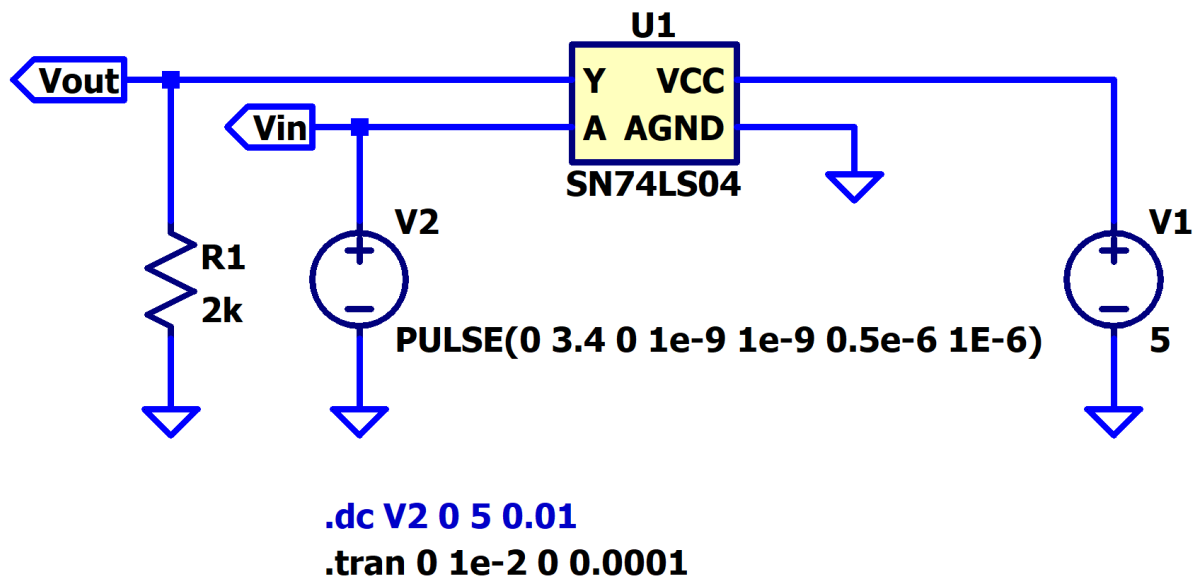
Frecuencia	Periodo	I_{prom} (μA)	Potencia (μW)
1 kHz	1 ms	740.83	124.16
10 kHz	100 μs	740.83	124.16
100 kHz	10 μs	740.83	124.16
500 kHz	2 μs	740.83	124.16
1 MHz	1 μs	740.83	124.16

la potencia estática se mantiene constante al subir la frecuencia ya que el modelo no considera la capacitancia interna del dispositivo sin embargo esta difiere al medirse en estado bajo y alto siendo de 124.16 μW y de 99.28 μW respectivamente.

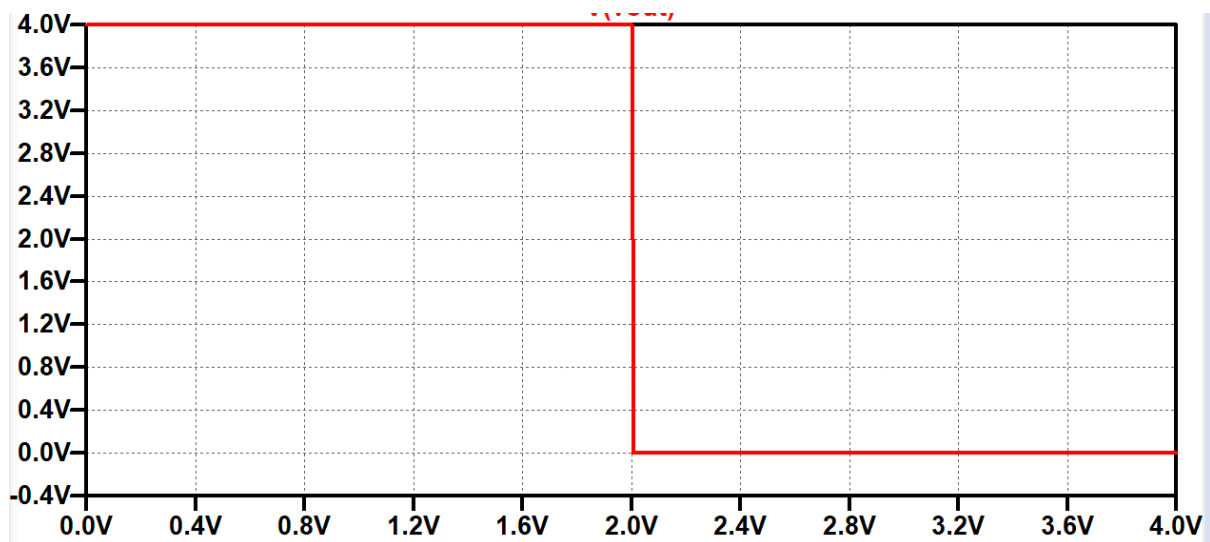
simulación de la potencia dinámica para el TTL.

<i>Frecuencia</i>	<i>Periodo</i>	$P_{din} (mW)$
1 kHz	1 ms	1.01
10 kHz	100 μs	1.02
100 kHz	10 μs	1.05
500 kHz	2 μs	1.22
1 MHz	1 μs	1.37

CMOS



A continuación tenemos la función de transferencia del CMOS



- Consumo de potencia simulado (comparar con los cálculos de la sección 3.2/3.3) → **"Ver Figura 3 en anexo"**

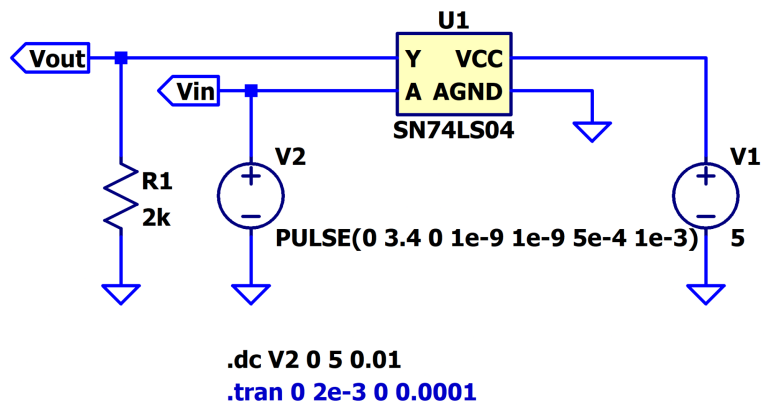
<i>Frecuencia</i>	<i>Periodo</i>	<i>Potencia (nW)</i>
1 kHz	1 ms	308.94
10 kHz	100μs	308.94
100 kHz	10μs	308.94
500 kHz	2μs	308.94
1 MHz	1μs	308.94

Simulación de potencia dinámica del CMOS

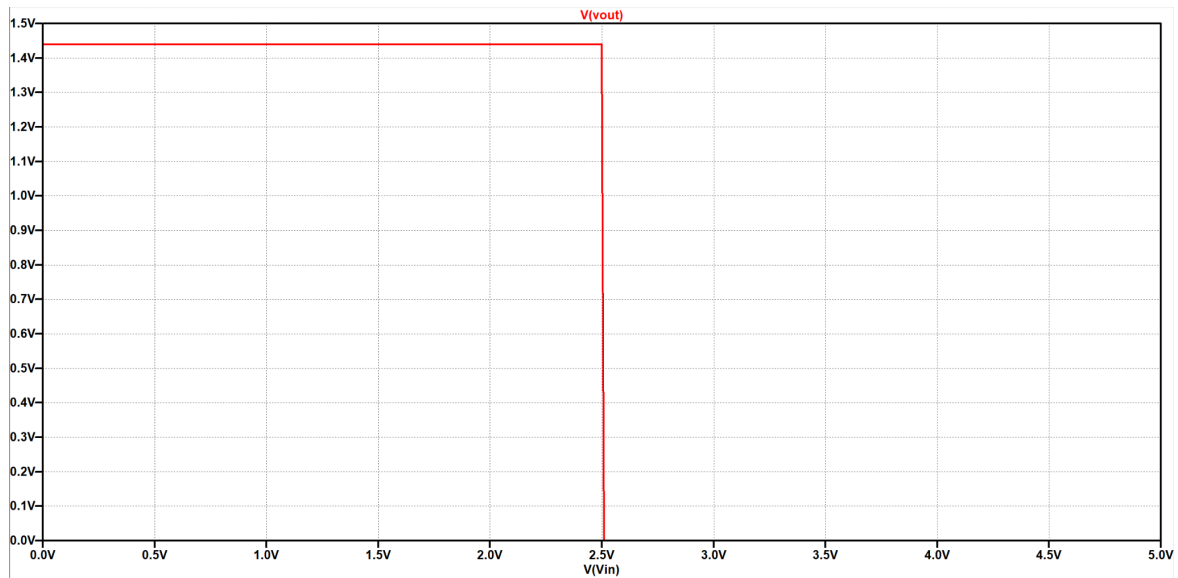
<i>Frecuencia</i>	<i>Periodo</i>	<i>Potencia (uW)</i>
1 kHz	1 ms	330.64

10 kHz	100µs	358.96
100 kHz	10µs	373.61
500 kHz	2µs	460.96
1 MHz	1µs	593.38

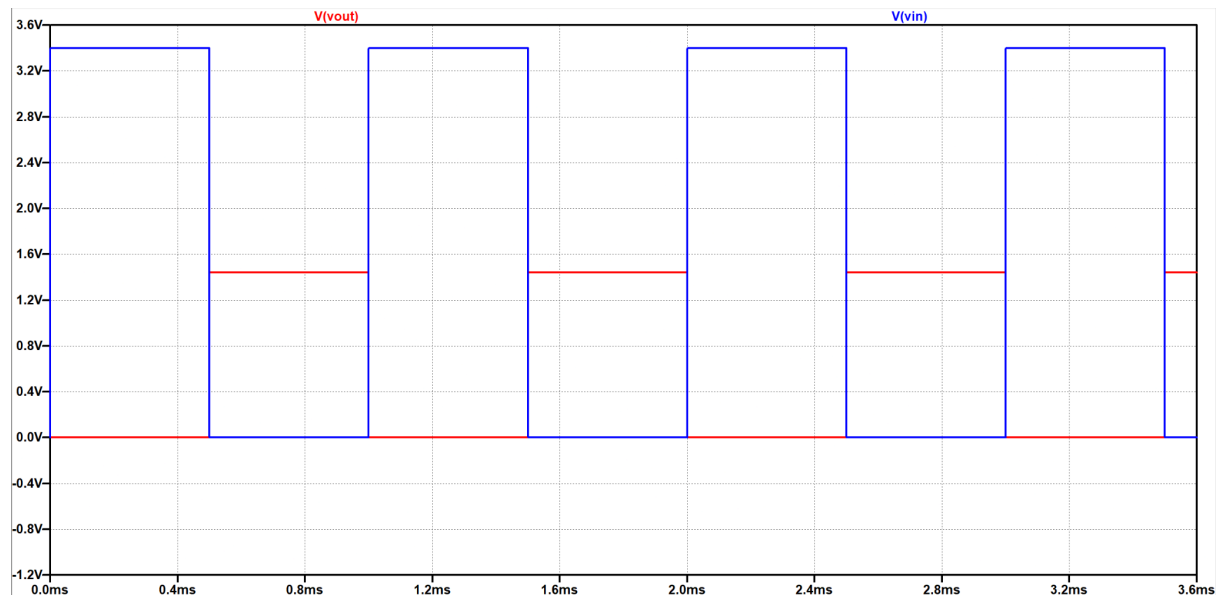
circuito modelado



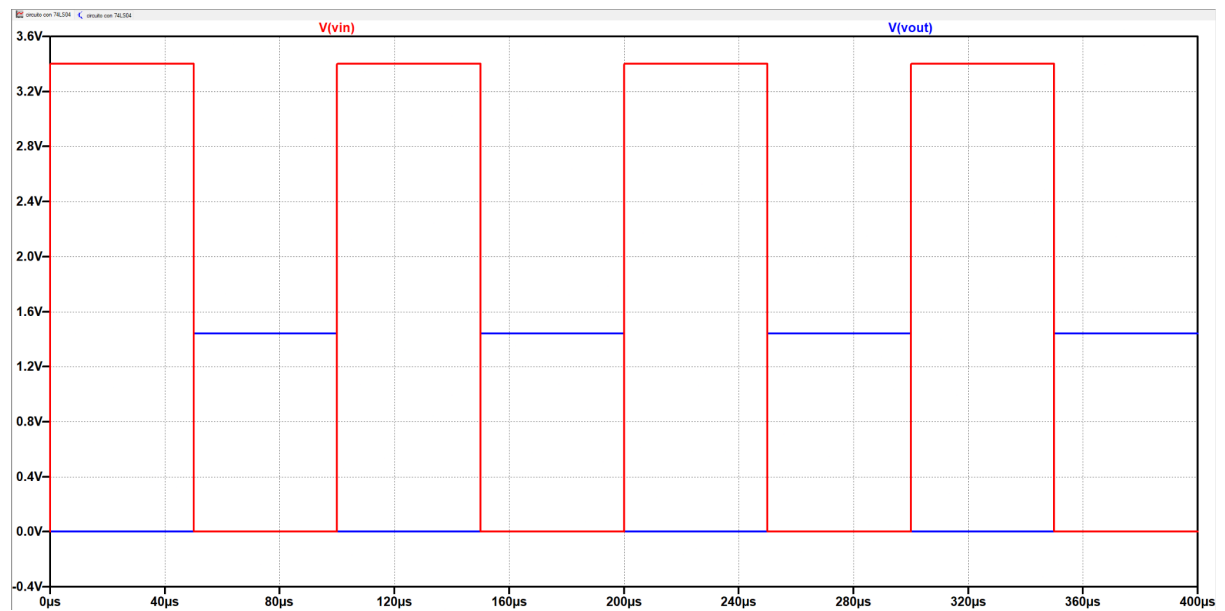
función de transferencia



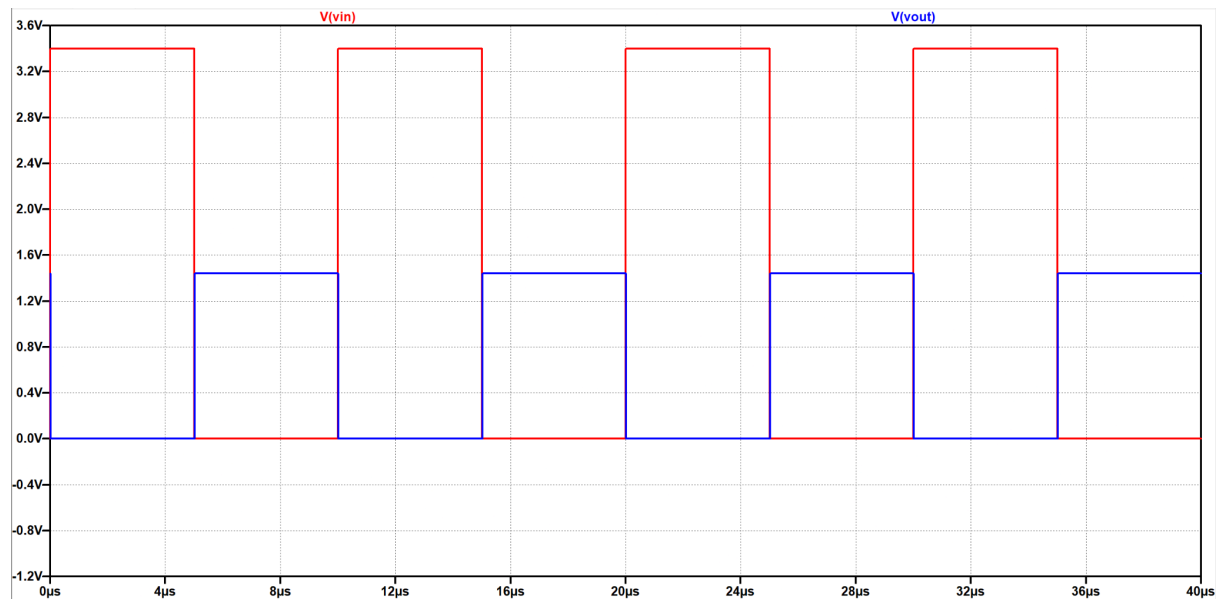
1KHz



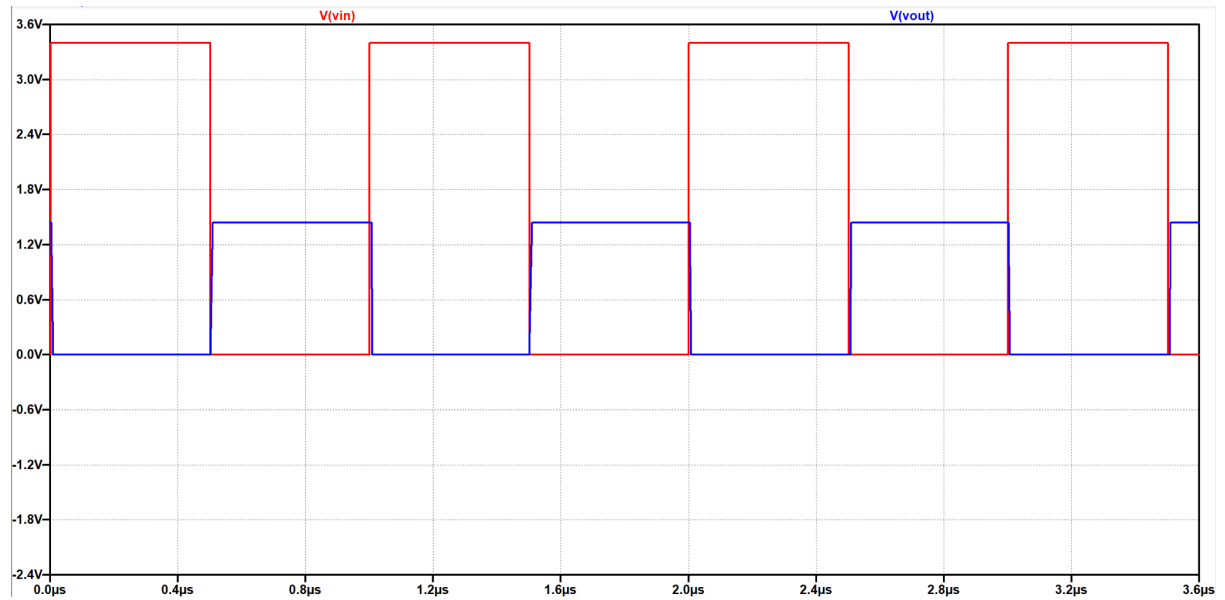
10KHz



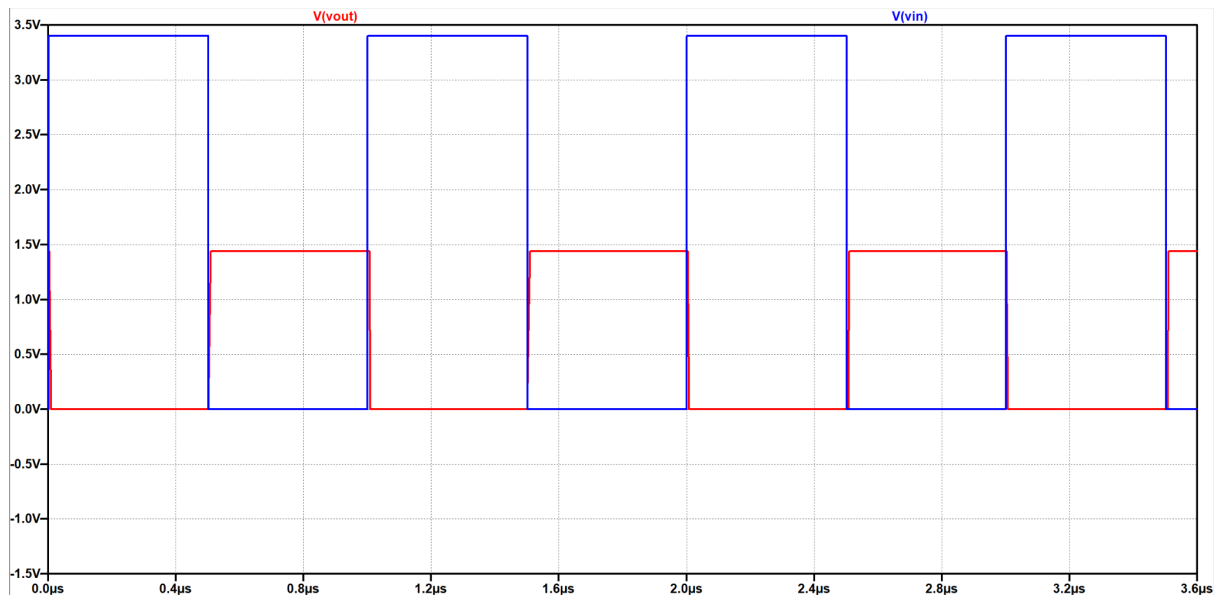
100KHz



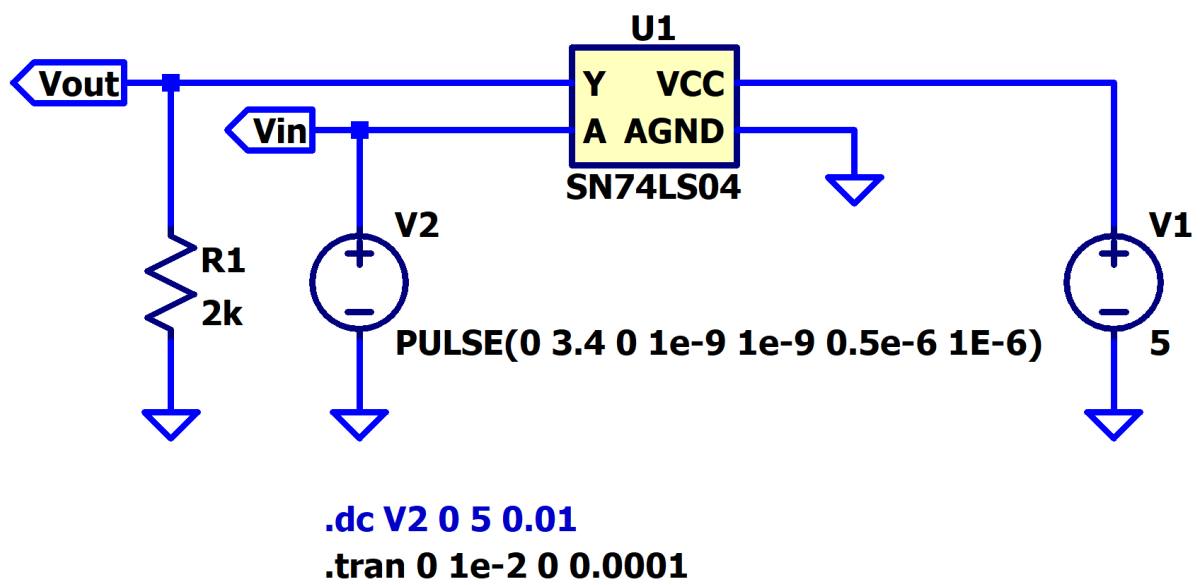
500 KHz



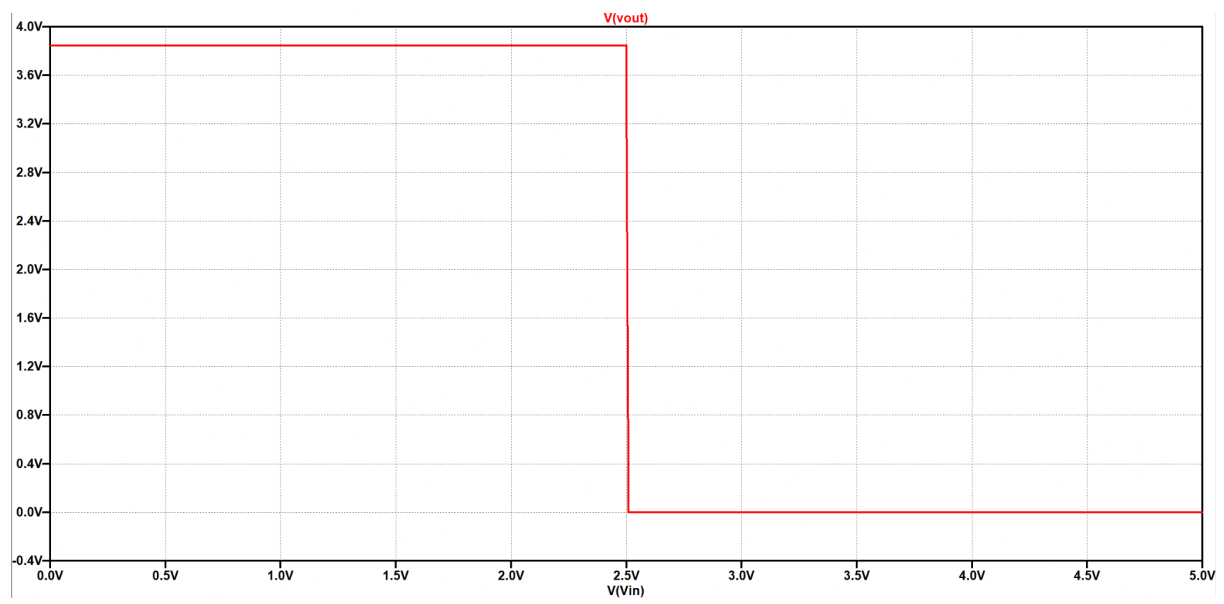
1 MHz



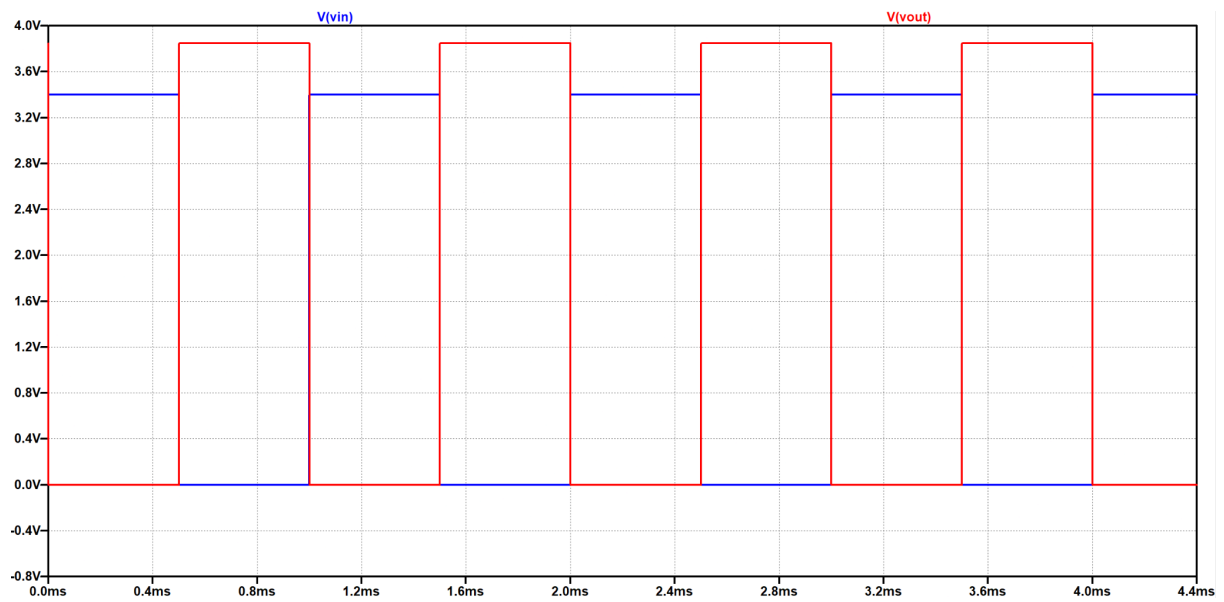
circuito modelado



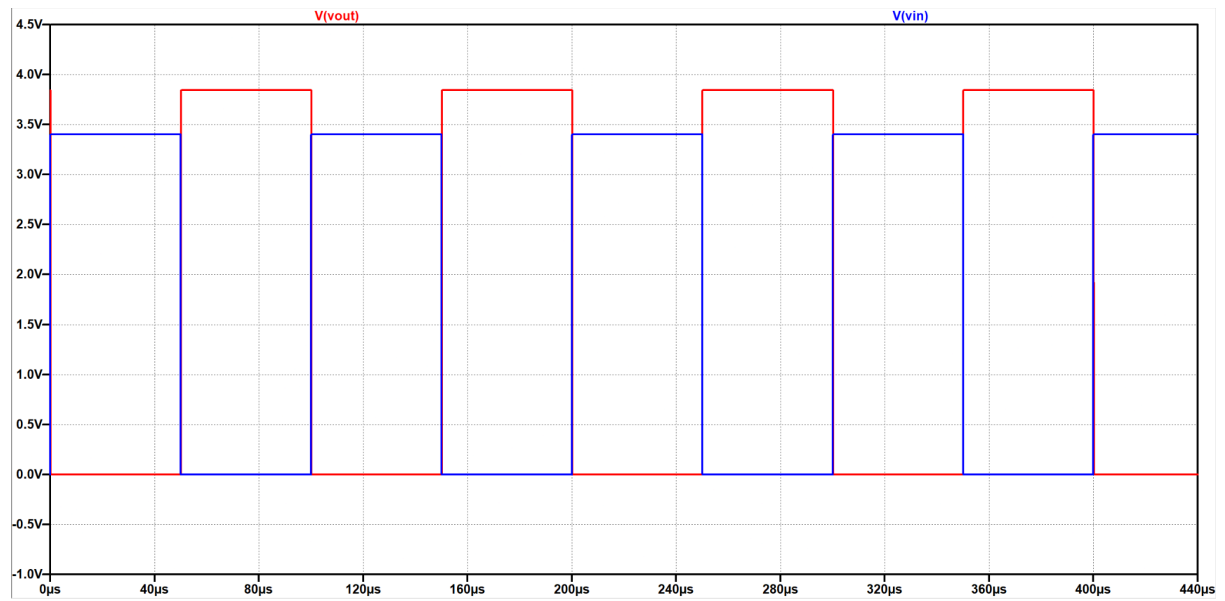
función de transferencia



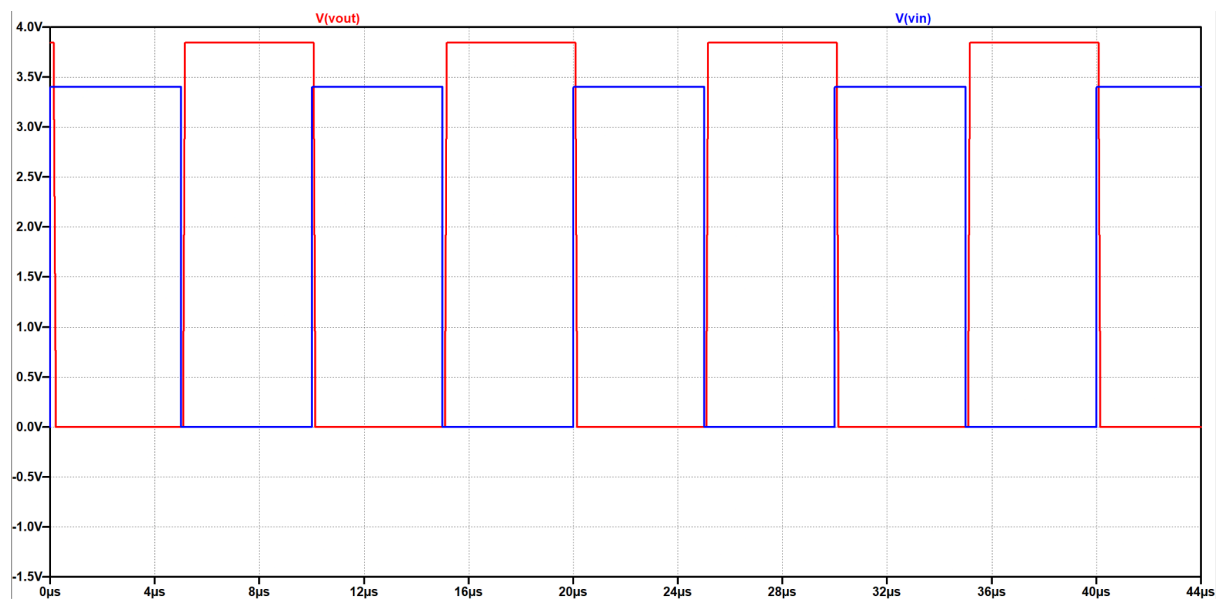
1KHz



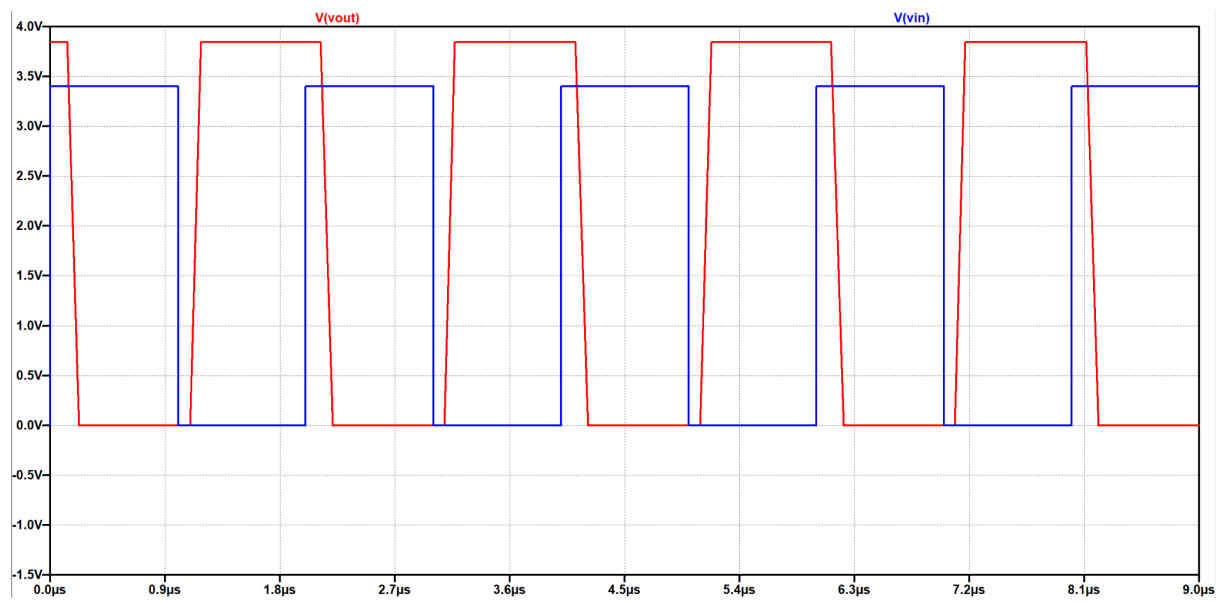
10KHz



100KHz



500 KHz



1 MHz

